

5th Grade Written Exam

2026 James Clemens Math Tournament

INSTRUCTIONS:

1. You will have 60 minutes to complete this examination.
2. This exam consists of 15 multiple choice questions and 3 tiebreaker questions. The multiple choice questions are each worth 4 points if answered correctly and -1 points if answered incorrectly. There is no penalty for unanswered questions. The tiebreaker questions are worth 0.1 points if answered correctly and 0 points if answered incorrectly or left unanswered.
3. Use a #2 pencil.
4. Calculators, cellular devices, books, notes, and other aids are prohibited during this exam. Scratch paper will be provided for calculations. If you require additional scratch paper, simply raise your hand and your proctor will assist you.
5. Mark your answers to the questions in the Scantron form provided. You may use the test booklets for scratch work, but only the answers marked in the Scantron form will be counted. Make sure to fill in the bubbles on the Scantron form **completely** and erase any stray marks.
6. Write your answers to the tiebreaker questions on the back of the Scantron form. Label the tiebreakers as TB1, TB2, and TB3.
7. NOTA denotes "None Of The Above."
8. Diagrams are not necessarily drawn to scale.
9. In the event of a tie, answers will be evaluated backwards starting from multiple choice question 15 to determine a winner.
10. Although this math tournament is intended to demonstrate your knowledge and skills in math, be sure to enjoy the experience and have fun!
11. You may keep the test booklet after the exam.

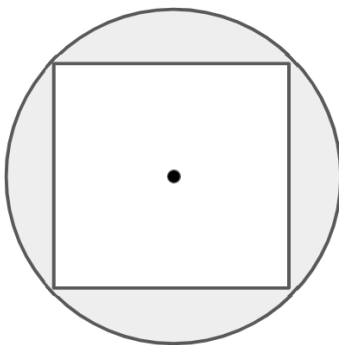
DO NOT TURN THIS PAGE UNTIL TOLD TO DO SO

1. Solve: $\frac{3}{4} + \frac{2}{7} - \frac{1}{8}$
(A) $\frac{65}{56}$ (B) $\frac{13}{28}$ (C) $\frac{51}{56}$ (D) $\frac{43}{56}$ (E) NOTA
2. An isosceles triangle has a base angle measuring 35° . What is the measure of its vertex angle?
(A) 110° (B) 145° (C) 72.5° (D) 35° (E) NOTA
3. Krish is having some trouble in his AP Physics class. His recent test scores are 68, 79, 91, 47, and 72. Which of the following scores can he NOT receive on his next test in order to get a "C" average? Note: A "C" average is a grade of 70 to 79, inclusive.
(A) 63 (B) 60 (C) 100 (D) 71 (E) NOTA
4. Solve for x : $3(4x + 5) - 4(6x + 3) = 39$
(A) 3 (B) -3 (C) 3.5 (D) -3.5 (E) NOTA
5. A rectangle has a perimeter of 40 units. What is the largest possible area?
(A) 64 (B) 80 (C) 96 (D) 100 (E) NOTA
6. Sid the Sloth is trying to climb up a 50 foot tall tree. Every 5 minutes, he climbs 4 feet, but falls asleep and drops 2.5 feet. How many minutes will it take for him to reach the top of the tree?
(A) 155 (B) 170 (C) 165 (D) 160 (E) NOTA
7. Rachel and Laura are making paper airplanes. Rachel can make 3 airplanes in 2 minutes, and Laura can make 5 airplanes in 3.5 minutes. If they work together, how many airplanes can they make in half an hour? Express your answer as a mixed number.
(A) $615\frac{5}{14}$ (B) $615\frac{1}{2}$ (C) $87\frac{5}{7}$ (D) $87\frac{6}{7}$ (E) NOTA
8. Jasmine has some cupcakes. She gives $\frac{1}{4}$ of them to her friend, Ella. She then gives $\frac{2}{3}$ of what is remaining to her two brothers and splits it evenly between them. She now has 14 cupcakes left. How many cupcakes did each brother receive?
(A) 28 (B) 7 (C) 12 (D) 14 (E) NOTA

9. Kelly has created a magic square with the numbers 1 – 9. She tells you that each row and each column sum to the same number. What is $A \cdot B - C \cdot D$?

8	A	6
B	5	C
4	D	2

- (A) 66 (B) -63 (C) -61 (D) 64 (E) NOTA
10. Let $f(x) = 5x + 2$ and $g(x) = 7x - 8$. What is $f(g(6))$?
- (A) 170 (B) 171 (C) 172 (D) 173 (E) NOTA
11. What is 642 written in base 8?
- (A) 1202_8 (B) 418_8 (C) 1308_8 (D) 532_8 (E) NOTA
12. Manny is watching the weather channel. There is a 60% chance of rain on Monday. If it rains on Monday, there is a 40% chance of rain on Tuesday. If it doesn't rain Monday, the chance that it rains on Tuesday increases to 55%. What is the probability that it does not rain on Monday but rains on Tuesday?
- (A) 0.22 (B) 0.16 (C) 0.24 (D) 0.55 (E) NOTA
13. A square is inscribed in a circle as shown in the figure below. If the diagonal of the square is 8 cm, what is the area of the shaded region?



- (A) 16π (B) $4\pi - 8$ (C) $16\pi - 32$ (D) $8\pi - 16$ (E) NOTA

14. Let A equal the number of zeros at the end of $21!$. Let B equal the probability of rolling a composite number using a fair six-sided die. Let C equal the number of positive integer factors of 132. What is $(A \cdot B)/C$?
- (A) $\frac{1}{12}$ (B) $\frac{1}{9}$ (C) $\frac{1}{6}$ (D) $\frac{1}{4}$ (E) NOTA
15. What is the units digit of $7^{37} + 2^{42} + 5^{27}$?
- (A) 8 (B) 6 (C) 0 (D) 2 (E) NOTA

Tiebreakers:

- TB1. Find the sum of the exponents in the prime factorization of 312.
- TB2. What is the 55th term in the following sequence: 1, 2, 2, 3, 3, 3, 4, 4, 4, 4, ...
- TB3. You have 15 coins made only of nickels and dimes worth \$1.05 total. What is the difference between the number of nickels and the number of dimes you have?